# ***Problem Statement:***

## FIFA 2022 Player Rating Analysis Using Hadoop Framework

The rapid growth of sports analytics has resulted in the generation of large-scale datasets containing detailed information about players, teams, and performance metrics. The FIFA 2022 dataset includes thousands of professional football players from various nationalities, with attributes such as overall rating, potential, position, club, market value, and wages. Analyzing such a vast dataset using traditional data processing techniques is inefficient, time-consuming, and lacks scalability.

The **primary problem addressed in this project is how to efficiently process and analyze large-scale FIFA player data to extract meaningful insights.** Specifically, the project aims to identify patterns in player ratings, determine top-performing nationalities, and understand the distribution of player performance across different regions. Handling such data involves challenges related to volume, processing speed, and structured analysis.

To overcome these challenges, the Hadoop ecosystem is utilized. **Hadoop Distributed File System (HDFS) enables distributed storage of large datasets, MapReduce facilitates parallel data processing across multiple nodes, and Hive allows SQL-like querying for efficient data analysis.** This combination ensures scalability, fault tolerance, and optimized performance in handling big data.

The objective of this project is to transform raw FIFA 2022 player data into actionable insights using big data technologies. By leveraging Hadoop, the project demonstrates how large datasets in sports analytics can be effectively processed to support better decision-making and reveal meaningful patterns in global football talent.